NUI Galway
OÉ Gaillimh

Semester 1 Examinations 2012/ 2013

Exam Code(s)	1MF1, 1SD1, 1MF2
Exam(s)	MSc in Software Design and Development, Higher Diploma in Applied Science (Software Design & Development),
Module Code(s)	CT853 Algorithmics & Logical Methods
Module(s)	Algorithmics & Logical Methods
Paper No.	
Repeat Paper	No
Discipline(s)	Information Technology
Internal Examiner(s)	Professor G Lyons Dr. M Madden Dr. Enda Howley* Dr. Conn Mulvihill*
External Examiner(s)	Professor M. O'Boyle
No. of Pages	2
Duration	**2 hours**
Instructions:	Attempt 4 questions in total Answer atleast **ONE** from Section A and Section B. All questions will be marked equally.

Requirements:
MCQ
Handout
Statistical/ Log Tables
Cambridge Tables
Graph Paper
Log Graph Paper
Other Materials

Release to Library: Yes

PTO

Section A

Question 1

(a) Explain with the aid of examples each of the following terms:
- HIPO Charts
- Event Driven Programming
- Big O Notation
- Pseudocode
- Recursion

(15 marks)

(b) Explain the meaning of operator precedence and make reference to its effects when calculating the following:

```
3 * (6 + 2) / 12 - (7 - 5) ^ 2 * 3
```

(10 marks)

Question 2

(a) Bubble and Quick Sort are two commonly discussed sorting algorithms. With the aid of examples and pseudocode you are asked to outline their features and structure.

(15 marks)

(b) You are now asked to discuss these two algorithms with respect to their Big O Notation.

(10 marks)

Question 3

(a) Given an integer array of size N with values in the range [0-10], write an algorithm using pseudocode which counts the frequencies of the integers (i.e., the number of times each integer occurs) and outputs the frequency for each integer which occurs at least once. For example, for the array with values:

2, 4, 10, 1, 10, 3, 1, 0, 1, 2

The output should be:

Frequency of number 0 is 1
Frequency of number 1 is 3
Frequency of number 2 is 2
Frequency of number 3 is 1
etc.

(5 marks)

(b) Given an integer array of size N, containing N values, write an algorithm in pseudocode to calculate, and output, the average of the N values.

(5 marks)

(c) Partial Java code for insertion sort is shown below. You are asked to complete the code and illustrate how the final code would work on a sample sorting task of sample numbers: 10, 1, -3, 39, 8, -17, 13, 5. Each step of the process should be clearly shown and explained.

(15 marks)

```
void insertionSort(int[] arr) {

      int i, j, newValue;

            for (i = 1; i < arr.length; i++) {

                 newValue = arr[i];

                 j = i;

                  while (j > 0 && arr[j - 1] > newValue) {

                  }

                        arr[j] = newValue;

            }

}
```

Section B

Question 4

(a)
A program is required to find the length of the longest plateau in the array b[0..N-1], N>0. The array contains positive integers. A plateau is taken to mean a sequence of adjacent and identical values. With an array that contains 4,5,5,5,6,6,6,6,7,2 the program should report '4' as the length of the longest plateau. Develop such a program in pseudocode (15 marks)

(b)
A program is required to perform a binary search on the array b[0..N-1], N>0. The array contains positive integers. You may assume that the array is ordered by <=. Develop such a program in pseudocode, explaining what is meant by the term 'binary search' in the course of your answer (10 marks)

Question 5

In the graph G, A and B are connected, A and C are connected, B and D are connected, B and E are connected, C and D are connected, C and F are connected, D and G are connected, E and G are connected, F and G are connected.

(a) Outline in pseudocode an algorithm for a breadth-first search of this graph, starting at A and seeking the goal G (15 marks)

(b) Given the following known costs for edges in G:
A-B : 1, A-C: 12, B-D: 1, B-E: 40, C-D:25, C-F:14, D-G:4, E-G:30, F-G:30
Explain what is meant by a 'uniform cost search' and outline how a uniform cost search would proceed by sketching in its entirety a uniform cost search tree starting at A and seeking the goal G. (N.B. It is <u>not</u> required to develop an algorithm in pseudocode; just develop a full trace of the appropriate search tree) (10 marks)

Question 6

(a)In the context of program development, explain what is meant by the following terms: Precondition, postcondition, invariant (10 marks)

(b)
Consider the postcondition P:
P: (0<=a*a) AND (a*a <= n) AND (n < (a+1)(a+1))
Suggest what a program that satisfies this postcondition P is attempting to calculate and develop a suitable invariant for this program (15 marks)